

Homework 8

Zhiqi Gan

Problem 8.1. (More Inclusion-Exclusion Principals).

1. Prove that for subsets X, Y, Z we have $|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$.
2. Find an example of subspaces X, Y, Z such that $\dim(X + Y + Z) \neq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$.
3. Prove that for subspaces X, Y, Z we have $\dim(X + Y + Z) \leq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$. (Hint: Use the regular IEP for subspaces repeatedly. Prove that $(X \cap Z) + (Y \cap Z) \subseteq (X + Y) \cap Z$).

Solution 8.1.

1. Proof:

Note that $|X|$ means the number of elements in X .

Method 1:

By Venn Diagram, in $|X| + |Y| + |Z|$, we count $|X \cap Y|, |X \cap Z|, |Y \cap Z|$ twice and count $|X \cap Y \cap Z|$ three times. So we need to subtract $|X \cap Y|, |X \cap Z|, |Y \cap Z|$. Now we have $|X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z|$. But now we subtract $|X \cap Y \cap Z|$ three times, we have to add it to make sure every element we count exactly once. Hence $|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$.

Method 2:

Let's prove two lemma first.

Lemma 1: $|X \cup Y| = |X| + |Y| - |X \cap Y|$.

We define $X \setminus Y$ as a set containing all elements in X but not in Y . Therefore,

$$|X \cup Y| = |X \setminus Y| + |Y| \quad (1)$$

$$|X| = |X \setminus Y| + |X \cap Y| \quad (2)$$

From (1) and (2), we immediately have

$$|X \cup Y| = |X| + |Y| - |X \cap Y|$$

Alternatively, you can use Venn Diagram to prove this.

Lemma 1 proof ends.

Lemma 2: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$.

The proof is trivial using Venn Diagram.

Lemma 2 proof ends.

So

$$\begin{aligned} |X \cup Y \cup Z| &= |(X \cup Y) \cup Z| \\ &= |X \cup Y| + |Z| - |(X \cup Y) \cap Z| \\ &= |X| + |Y| + |Z| - |X \cap Y| - |(X \cup Y) \cap Z| \\ &= |X| + |Y| + |Z| - |X \cap Y| - |(X \cap Z) \cup (Y \cap Z)| \\ &= |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z| \end{aligned}$$

2. $X = \text{span}(\mathbf{e}_1), Y = \text{span}(\mathbf{e}_2), Z = \text{span}(\mathbf{e}_1 + \mathbf{e}_2)$. And the answer is not unique.

3.Proof:

$$\begin{aligned} \dim(X + Y + Z) &= \dim(X + Y) + \dim(Z) - \dim((X + Y) \cap Z) \\ &= \dim(X) + \dim(Y) - \dim(X \cap Y) + \dim(Z) - \dim((X + Y) \cap Z) \end{aligned}$$

$\forall \mathbf{x} \in (X \cap Z) + (Y \cap Z), \mathbf{x} = \mathbf{m} + \mathbf{n}$, where $\mathbf{m} \in (X \cap Z)$ and $\mathbf{n} \in (Y \cap Z)$. Since Z is a subspace, and $\mathbf{m}, \mathbf{n} \in Z$, we have

$$\mathbf{x} = (\mathbf{m} + \mathbf{n}) \in Z \quad (3)$$

Similarly, $\mathbf{m} \in X, \mathbf{n} \in Y$, so

$$\mathbf{x} = \mathbf{m} + \mathbf{n} \in (X + Y) \quad (4)$$

From (3) and (4), $\mathbf{x} \in ((X + Y) \cap Z)$. That is to say, $\forall \mathbf{x} \in (X \cap Z) + (Y \cap Z)$, we also have $\mathbf{x} \in ((X + Y) \cap Z) \implies (X \cap Z) + (Y \cap Z) \subseteq ((X + Y) \cap Z) \implies \dim((X \cap Z) + (Y \cap Z)) \leq \dim((X + Y) \cap Z)$.

Therefore,

$$\begin{aligned}\dim(X + Y + Z) &= \dim(X) + \dim(Y) - \dim(X \cap Y) + \dim(Z) - \dim((X + Y) \cap Z) \\ &\leq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim((X \cap Z) + (Y \cap Z)) \\ &= \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)\end{aligned}$$

Problem 8.2.

Solution 8.2.

1. Set $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_d]$, by definition of Range and basis, $\text{Ran}(B) = \text{span}(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_d) = V \implies \text{Ker}(B^T) = \text{Ran}(B)^\perp = V^\perp$. Since column vectors of B are basis $\implies$ they are independent. $\implies B$ is injection. $\implies \text{ker}(B) = \{\mathbf{0}\}$.

2. Same as 1.

3. From 1, 2, we have $\text{ker}(B) = \text{ker}(C) = \{\mathbf{0}\}$, then $CB^T \mathbf{x} = C(B^T \mathbf{x}) = \mathbf{0} \iff B^T \mathbf{x} = \mathbf{0}, BC^T \mathbf{x} = B(C^T \mathbf{x}) = \mathbf{0} \iff C^T \mathbf{x} = \mathbf{0}$.

4. From 3 and 1, we have

$$\text{ker}(CB^T) = \text{ker}(B^T) = V^\perp, \quad \text{ker}((CB^T)^T) = \text{ker}(BC^T) = \text{ker}(C^T) = W^\perp.$$

Then,

$$\text{Ran}(CB^T) = \text{ker}((CB^T)^T)^\perp = (W^\perp)^\perp = W.$$

$$\text{Ran}((CB^T)^T) = \text{ker}(CB^T)^\perp = (V^\perp)^\perp = V.$$

Hence CB^T is the desire matrix.

Problem 8.3.

Solution 8.3.

1. The augmented matrix of $A\mathbf{x} = \mathbf{b}$ is $\left[\begin{array}{ccc|c} 1 & 2 & 2 & 5 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 & 9 \end{array} \right]$, then

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 5 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 & 9 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 2 & 5 \\ 0 & -2 & -1 & -5 \\ 0 & -2 & -1 & -6 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 2 & 5 \\ 0 & -2 & -1 & -5 \\ 0 & 0 & 0 & -1 \end{array} \right]$$

So $A\mathbf{x} = \mathbf{b}$ has no solution.

The augmented matrix of $\begin{bmatrix} A^T \\ \mathbf{b}^T \end{bmatrix} \mathbf{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$ is $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 2 & 4 & 0 \\ 2 & 3 & 5 & 0 \\ 5 & 5 & 9 & 1 \end{array} \right]$, then

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 2 & 4 & 0 \\ 2 & 3 & 5 & 0 \\ 5 & 5 & 9 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -5 & -6 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So the solution is $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

2.

$$(x + 2y + 2z) + (2x + 2y + 3z) - (3x + 4y + 5z) = 5 + 5 - 9 \implies 0 = 1$$

3.

Proof:

First, we prove that $A\mathbf{x} = \mathbf{b}$ has a solution. Since $\mathbf{b} \in \text{Ran}(A)$, then $\mathbf{b}$ is a linear combination of column vectors of

A , so $A\mathbf{x} = \mathbf{b}$ has a solution.

Next, we prove that $\begin{bmatrix} A^T \\ \mathbf{b}^T \end{bmatrix} \mathbf{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$ has no solution.

Assumed that $\mathbf{y}$ is a solution of the equation, then $A^T \mathbf{y} = \mathbf{0}, \mathbf{b}^T \mathbf{y} = 1 \implies \mathbf{y} \in \ker(A^T)$. Since $\ker(A^T) = \text{Ran}(A)^\perp$, then $\mathbf{y} \in \text{Ran}(A)^\perp \implies \mathbf{y} \perp \text{Ran}(A)$, which means $\mathbf{y}$ is perpendicular to all vectors in $\text{Ran}(A)$. However, we also have $\mathbf{b} \in \text{Ran}(A)$ and $\mathbf{b}^T \mathbf{y} = 1 \neq 0$. Contradiction!!! So the equation has no solution.

4.

Proof:

$\mathbf{b} \notin \text{Ran}(A) \implies A\mathbf{x} = \mathbf{b}$ has no solution.

Let's say the augmented matrix $[A \ \mathbf{b}]$ turn into its RREF after a collection of row operations E , that is, $E[A \ \mathbf{b}] = \text{RREF}([A \ \mathbf{b}])$. Because $A\mathbf{x} = \mathbf{b}$ has no solution, the last non-zero row (let's say it's the i -th row) of $\text{RREF}([A \ \mathbf{b}])$ is $\begin{bmatrix} \mathbf{0}^T & 1 \end{bmatrix}$. Thus,

$$\begin{aligned} \mathbf{e}_i^T \text{RREF}([A \ \mathbf{b}]) &= \begin{bmatrix} \mathbf{0}^T & 1 \end{bmatrix} \implies \mathbf{e}_i^T E[A \ \mathbf{b}] = \begin{bmatrix} \mathbf{0}^T & 1 \end{bmatrix} \\ &\implies \begin{bmatrix} A^T \\ \mathbf{b}^T \end{bmatrix} E^T \mathbf{e}_i = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} \\ &\implies \mathbf{y} = E^T \mathbf{e}_i \end{aligned}$$

Since the RREF of the matrix always exists, $\mathbf{y} = E^T \mathbf{e}_i$ always exists, too. The equation has a solution.

Problem 8.4.

Solution 8.4.

$$1. \text{ Set } A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \text{ then } \ker(A^T) = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}\right). \text{ So } \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}\right)^\perp = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}\right).$$

$$2. \begin{bmatrix} 2 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ 0 \\ 6 \\ -1 \end{bmatrix}.$$

$$3. \text{ A basis for the orthogonal complement is } \begin{bmatrix} b \\ -a \\ 0 \end{bmatrix}, \begin{bmatrix} c \\ 0 \\ -a \end{bmatrix}.$$

4.

It's obvious to see that we can use the first two rows to kill the last two rows, and the first two rows are linearly independent. So $\text{rank}(A) = 2$ because rank of A is the number of independent row vectors.

$$\text{Since } \text{rank}(A) = \dim \text{Ran}(A) = 2, \text{ and the first two columns are linearly independent, a basis for } \text{Ran}(A) \text{ is } \begin{bmatrix} r \\ p \\ 0 \\ 0 \\ 0 \\ p \\ r \end{bmatrix}, \begin{bmatrix} n \\ p \\ 0 \\ 0 \\ 0 \\ p \\ n \end{bmatrix}.$$

Alternatively, you can pick any two independent columns of A as a basis.

$$\text{A basis for } \text{Ran}(A^T): \begin{bmatrix} r \\ n \\ b \\ q \\ k \\ b \\ n \\ r \end{bmatrix}, \begin{bmatrix} p \\ p \\ p \\ p \\ p \\ p \\ p \\ p \end{bmatrix}.$$

By Rank-nullity Theorem, $\dim \ker(A^T) = \dim \ker(A) = 8 - \text{rank}(A) = 6$. Thus we expect 6 vectors in the basis of

these two subspace.

By observation, $\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5, \mathbf{e}_6, \mathbf{e}_1 - \mathbf{e}_8, \mathbf{e}_2 - \mathbf{e}_7$ are all in $\ker(A^T)$ and they are linearly independent. We've already found 6 independent vectors in the subspace, so they are basis of $\ker(A^T)$. Alternatively, you can also use the brute-force method and find the basis by RREF.

By Gaussian Elimination, the RREF of A is

$$\begin{bmatrix} 1 & 0 & \frac{b-n}{r-n} & \frac{q-n}{r-n} & \frac{k-n}{r-n} & \frac{b-n}{r-n} & 0 & 1 \\ 0 & 1 & \frac{r-b}{r-n} & \frac{r-q}{r-n} & \frac{r-k}{r-n} & \frac{r-b}{r-n} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Then a basis for $\ker(A)$ is

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} b-n \\ r-b \\ n-r \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} q-n \\ r-q \\ 0 \\ n-r \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} k-n \\ r-k \\ 0 \\ 0 \\ n-r \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} b-n \\ r-b \\ 0 \\ 0 \\ 0 \\ n-r \\ 0 \\ 0 \end{bmatrix}.$$

5.

Proof:

$$\text{Ran}(A^T) = \text{Ran}(B^T) \implies \text{Ran}\left(\begin{bmatrix} I \\ F^T \end{bmatrix}\right) = \text{Ran}\left(\begin{bmatrix} I \\ G^T \end{bmatrix}\right) \implies \text{each column of } \begin{bmatrix} I \\ G^T \end{bmatrix} \text{ is a linear combination of column}$$

$$\text{vector of } \begin{bmatrix} I \\ F^T \end{bmatrix} \implies \exists \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^n, \text{ such that } \begin{bmatrix} I \\ G^T \end{bmatrix} = \begin{bmatrix} I \\ F^T \end{bmatrix} [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_n] \implies \exists X \in \mathbb{R}^{n \times n} \text{ such}$$

$$\text{that } \begin{bmatrix} I \\ F^T \end{bmatrix} X = \begin{bmatrix} I \\ G^T \end{bmatrix} \implies IX = I, F^T X = G^T \implies X = I \implies F = G.$$

Problem 8.5.

Solution 8.5.

1. The rank of M is 3. 2. $X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ -1 & 1 \\ -\frac{1}{3} & 1 \\ -1 & 1 \end{bmatrix}$. 3. $Y = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$.

4.

$$M - XY = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{3} & -\frac{2}{3} & \frac{4}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

5. All columns of X are linearly independent, all rows of Y are linearly independent. And from 4, we have

$$M = XY + R, \text{ where } R \text{ is a rank-one matrix. Set } X = [\mathbf{x}_1 \ \mathbf{x}_2], Y = \begin{bmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \end{bmatrix}, \text{ then}$$

$$M = XY + R = [\mathbf{x}_1 \ \mathbf{x}_2] \begin{bmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \end{bmatrix} + R = \mathbf{x}_1 \mathbf{y}_1^T + \mathbf{x}_2 \mathbf{y}_2^T + R$$

The first rank-one matrix $\mathbf{x}_1 \mathbf{y}_1^T$ reflects how educational videos contributes to M , $\mathbf{x}_2 \mathbf{y}_2^T$ reflects how funny videos contributes to M , and R is related to other factors.